

Tufts MS in Computer Science Requirements (MSCS)



Total Requirements

- **10 total courses** (each must be 3 SHUs or more)
 - Must maintain a **minimum GPA of B**, with **no more than one grade below B-**
 - Courses numbered **below 100 do not count** for graduate credit
 - Graduation application required:
[Graduate Student Graduation Info](#)
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Core Competency Areas (1 course per area required)

At least one course must be completed in **each of the four areas** below, with a **minimum grade of B-**. Courses can be taken at Tufts or transferred with transcript evidence from another university.

1. Computer Architecture & Assembly Language (CA&AL)

- CS 111: Operating Systems
- CS 112: Networks
- CS 114: Network Security
- CS 116: Introduction to Security
- CS 118: Cloud Computing
- CS 140: Advanced Topics in Computer Architecture
- CS 146 (also EE 126): Computer Engineering
- CS 107 (Formerly COMP 181): Compilers (*offered infrequently*)

2. Programming Languages (PL)

- CS 105: Programming Languages
- CS 121 (Formerly COMP 180): Software Engineering
- CS 107 (Compilers) (*also satisfies CA&AL*)

3. Data Structures & Analysis of Algorithms (DS&AA)

- **CS 160: Intro to Algorithms** (*strongly recommended*)

4. Theory of Computation (ToC)

- CS 170: Computation Theory
 - Suggested: COMP 61 (Discrete Math) if lacking math background
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Research & Project Options

CS 191 – Research (1 time only)

- Counts as 1 research course
- Not to be taken more than once

CS 199 – Internship Course (for international students)

- **Does not count** towards 10-course requirement
 - For CPT/internship authorization (contact Prof. Ming Chow)
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M.S. Project Option

- **Register for CS 293**, usually 3 SHUs
 - Faculty-supervised research project without writing a full thesis
 - Must be planned and approved by faculty advisor
 - May be taken **only once**
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M.S. Thesis Option

- **Two-semester commitment:** enroll in **CS 295** and **CS 296** (any order)
 - Requires finding a faculty advisor and switching to thesis track
 - Must submit:
 - **Thesis Prospectus** (after first semester)
 - **Final Thesis Document**
 - **45-minute public defense**
 - Committee:
 - At least **3 faculty members**, with **1 outside CS department**
 - Grade for CS 295 & 296 given **after both semesters completed**
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Directed Study Courses (CS 193-xx, xx > 01)

- Faculty-designed group reading/implementation/study
 - **Count as standard courses**
 - Unlimited use toward 10-course requirement
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Summary of Acceptable Course Types

Type	Course Codes	Notes
Standard Course	CS xxx	Main lecture-based courses
Research Course	CS 191, 293, 295, 296, 297	Restrictions on repeats
Directed Study	CS 193-xx (xx > 01)	Count as standard lecture courses
Internship	CS 199	Does not count towards 10 total



Academic Standing

- GPA $\geq$ **B average**
- No more than **1 grade below B-**
- Continuous progress required